COMSATS University Islamabad,
Sahiwal Campus.

CUI Chathub

A Project Presented To

COMSATS University Islamabad,
Sahiwal Campus

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Software Engineering (2020-2024)

By

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The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of other

DECLARATION

We hereby declare that this software, neither whole nor as a part has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely based on our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Ahmad Ali	**Muhammad Mumtaz**	**Rana Muhammad Zuhair**

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS(SE) **"CUI Chathub"** was developed by **Ahmad Ali (CIIT/SP20-BSE-004/SWL)**, **Muhammad Mumtaz (CIIT/SP20-BSE-015/SWL)** and **Rana Muhammad Zuhair (CIIT/SP20-BSE-021/SWL)** under the supervision of "**Ms. Madiha Fatima**" and co supervisor "**Mr. Ahmad Shaff**" and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Software Engineering.

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Executive Summary

CUI Chathub is a distinctive chat application that caters exclusively to university students. Unlike generic chat platforms, CUI Chathub has been specifically designed to meet the unique challenges and requirements of university life. This exclusive platform allows students to create and join groups, enabling connections with peers who share their academic activity or personal interests. Real-time chat capabilities simplify communication, making it a valuable tool for academic collaborations and social interactions.

The user-friendly interface ensures that students of all technical backgrounds can easily navigate the application. For university societies, CUI Chathub provides a streamlined way to share announcements, ensuring that their activities and updates reach the right audience.

The app's development will follow an agile model, allowing for the implementation of the features in stages. The development process will be guided by the functional requirements of the app. The app's non-functional requirements will be considered to ensure its usability, reliability, and security.

The app's testing and evaluation will involve unit testing, integration testing, and system testing to ensure that all functionalities are working as intended. The app will also undergo user acceptance testing to gather feedback from users and improve the app's usability.

In summary, CUI Chathub provides a dedicated space for productive and engaging conversations within the university community. This innovative platform enhances connectivity, knowledge sharing, and overall university experiences, making it an essential resource for students seeking to maximize their educational journey.

Acknowledgment

All praise is to almighty Allah who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor **"Ms. Madiha Fatima"** and our Co-Supervisor **"Mr. Ahmad Shaff".** Without their supervision, advice, and valuable guidance, the completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

Ahmad Ali **Muhammad Mumtaz** **Rana Muhammad Zuhair**

__________________ __________________ __________________

Abbreviations

API	Application Programming Interface
JS	JavaScript
SRS	Software Require Specification
SDD	Software Design Description
GUI	Graphical User Interface
IDEs	Integrated development environments
CSS	Cascading Style Sheet

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Chapter 1
Introduction

1 Introduction

CUI Chathub is a revolutionary addition to the university ecosystem, addressing the unique requirements and challenges that students face during their academic journey. This exclusive chat application has been carefully crafted for the students of the COMSATS University Islamabad (CUI), Sahiwal Campus. It is tailored to meet the specific needs of students on campus. This exclusivity creates environment where students can connect with their peers and engage in real time chat, whether they are in the same department, share common interests, or are part of the same university society. This exclusivity ensures that conversations remain relevant and meaningful, making it the ideal platform for university-related discussions.

Furthermore, CUI Chathub serves as a vital tool for university societies. These organizations often struggle to disseminate information and announcements effectively. With this app, they can easily reach their target audience, ensuring that the right people receive their activities, events, and updates. This feature streamlines communication and strengthens the sense of belonging within the university community.

Currently, there is no dedicated chat platform available for students of the COMSATS University Islamabad (CUI), Sahiwal Campus. Students face difficulties in reaching out other fellows with in the university. The CUI Chathub project aims to address this issue by providing a comprehensive and reliable chatting application for the users.

The user friendly interface is another key attribute of CUI Chathub. It ensures that students, regardless of their technical proficiency, can easily navigate the platform. Searching for and joining groups of interest is straightforward, and the app's intuitive design ensures that students can fully harness its capabilities without a steep learning curve. This user-friendliness makes it a valuable resource for both tech-savvy students and those who may not be as familiar with digital platforms.

In essence, CUI Chathub features to the unique needs and demands of university life. It provides an exclusive space where students can engage in boundary-free conversations that are both productive and engaging.

1.1 Brief Overview

The CUI Chathub project aims to develop a web application exclusively for the COMSATS University Islamabad, Sahiwal Campus users to connect them socially. The app will be developed using the MERN stack, a popular and robust platform for building high-performance web-based applications. The project will follow the agile software development methodology, allowing for iterative and systematic development. [1]

CUI Chathub is a unique and focused chat application designed to provide students at the university with a platform to engage in open conversations without constraints. While many chat apps exist for global communication, CUI Chathub's distinctive feature is its exclusivity to university users. It overcomes the challenges and difficulties faced in university life by enabling students to create groups, engage in discussions on various topics, both academic and personal.

1.2 Relevance to Course Modules

The development of CUI Chathub directly applies and integrates various concepts and skills acquired throughout our software engineering course.

i. Technical Skills:

Application of programming languages, frameworks, and tools learned during the course, including web development and database management.

ii. Software Engineering Principles:

Adherence to software engineering principles, encompassing software design, development methodologies, testing procedures, and project management. This subject influenced requirement analysis, system design, coding, testing, and maintenance phases, ensuring a comprehensive software engineering perspective [2]

iii. Database Design:

A solid understanding of database management is critical for any management system. Knowledge in this area influenced the design and implementation of the database schema for storing and retrieving data efficiently across various modules of the system

iv. System Architecture:

Understanding software design principles and architectural patterns was crucial in crafting a scalable and maintainable system. These subjects influenced decisions related to the systems overall structure, module interdependencies, and adherence to best practices for robust software architecture.

v. UI/UX Design:

A user-friendly interface is top priority for the success of any system. Application of UI/UX design principles from human-computer interaction and user interface design courses facilitated the design of an intuitive and ergonomic user interface, For CUI Chathub

vi. Project Management:

This subject provided insights into project planning, execution, and control. Principles of project management guided the systematic development and timely delivery of the CUI Chathub.

vii. Networking

Additionally, the real-time chat functionality in CUI Chathub necessitates a deep understanding of networking and communication protocols

1.3 Project Background

The need for the CUI Chathub project arises from several factors and experiences that occurred prior to its inception. These factors can be summarized in the project's background as follows:

Prior to CUI Chathub, university students often faced challenges in connecting with their peers in a campus-centric manner. Generic chat applications did not provide a dedicated space for university-specific discussions and interactions, making it difficult for students to engage in meaningful conversations related to their academic and personal interests within the university community.

Also, University societies struggled to disseminate announcements and event details to their members. This led to a lack of engagement and awareness among students regarding the various extracurricular opportunities available on campus.

In response to these challenges and unmet needs, the CUI Chathub project was conceived. It aims to provide a dedicated, campus-centric platform that overcomes these previous limitations by offering students the tools and features required for efficient, real-time communication, group management, and information sharing. By addressing these issues, the project enhances the overall university experience and strengthens the sense of community among students.

1.4 Methodology and Software Lifecycle for This Project

CUI Chathub is a software project with a variety of requirements. It is important to use a methodology and software lifecycle that is flexible, collaborative, and emphasizes continuous improvement. Agile is a good choice for this project because it meets all of these criteria.

1.4.1 Rationale behind Selected Agile Model

Agile is a software development methodology that emphasizes flexibility, collaboration, and continuous improvement. It is well suited for complex projects with changing requirements. Agile also advocates for ongoing user collaboration, which in this context means actively involving university students throughout the development process. Their feedback and evolving requirements should guide the platform's continuous improvement. [3]

Agile is based on the following principles:

•**Iterative development:** Software is developed in cycles, with each cycle adding new functionality or improving existing functionality.

•**Incremental delivery:** Software is delivered to users in increments, with each increment adding new value.

•**Continuous feedback:** Users and stakeholders provide feedback on the software throughout the development process.

•**Adaptive planning:** The development team adapts the plan as needed to meet the needs of the users and the changing requirements.

Chapter 2

Problem Definition

2 Problem Definition

This chapter discuss the precise problem to be solved. It should extend to include the outcome.

2.1 Problem Statement

The "CUI Chathub" project aims to address the prevalent issue of limited digital interaction among students within our university campus and the absence of a centralized platform for university societies to effectively communicate and share updates with the student body. This system is being developed with the overarching goal of enhancing student engagement, fostering collaboration among students and societies, and creating a unified platform that keeps the university community well-informed about society activities.

Currently, there is no existing system within our university that combines the specific features and functionalities offered by "CUI Chathub," making it a novel solution tailored to our university's unique communication needs. The absence of a comparable system underscores the significance of this project in fulfilling a gap in our current digital communication landscape.

2.2 Deliverables and Development Requirements

The CUI Chathub project's deliverables and development requirements encompass a range of components to ensure the successful implementation of the platform. These include:

- **User Interface (UI) Design:** Develop an intuitive and user-friendly interface that allows students to easily navigate and use the application. Consider mobile responsiveness for a seamless experience on various devices.
- **Real-Time Chat Functionality:** Implement a robust real-time chat system that supports one-on-one and group conversations.
- **User Registration:** Create a secure user registration system with customizable user profile.
- **Group Creation:** Design groups for students to create and join based on academic interests, societies, or other affiliations.
- **Search and Discovery**: Implement a robust search functionality that enables users to discover and join groups, find peers.
- **Security and Privacy:** Prioritize the security and privacy of user data, including encryption of access control.
- **Scalability:** Ensure the platform can handle a growing user base by implementing scalable architecture and cloud-based solutions.

- **Cross-Platform Compatibility**: Developing "CUI Chathub" as a web-based application would inherently make it cross-platform compatible. Users can access it through web browsers on various devices without the need for platform-specific apps.
- **Documentation**: Create user manuals and developer documentation to aid users and future development efforts.
- **Testing and Quality Assurance:** Implement comprehensive testing procedures to identify and rectify bugs and issues, ensuring a seamless user experience.
- **User Training and Support:** Develop user-training materials and provide support channels to assist users with the application.
- **Regular Updates and Maintenance:** Plan for ongoing software maintenance and updates to address evolving user needs and emerging technologies.

These deliverables and development requirements collectively contribute to the successful creation and continued improvement of CUI Chathub, aligning with the project's objectives of enhancing communication, collaboration, and community engagement among university students.

2.3 Current System

As of now, our university lacks a centralized communication platform, leading to challenges in facilitating seamless interactions among students, faculty, and staff. The absence of a dedicated system results in fragmented information dissemination and limited networking opportunities. To address these challenges, the development of CUI Chathub is initiated.

Chapter 3

Requirement Analysis

3 Requirement Analysis

3.1 Use Case Diagrams

In this Use case diagrams, we show the behavior of our system. This enable you to visualize the different types of roles in a system and how those roles interact with the system.

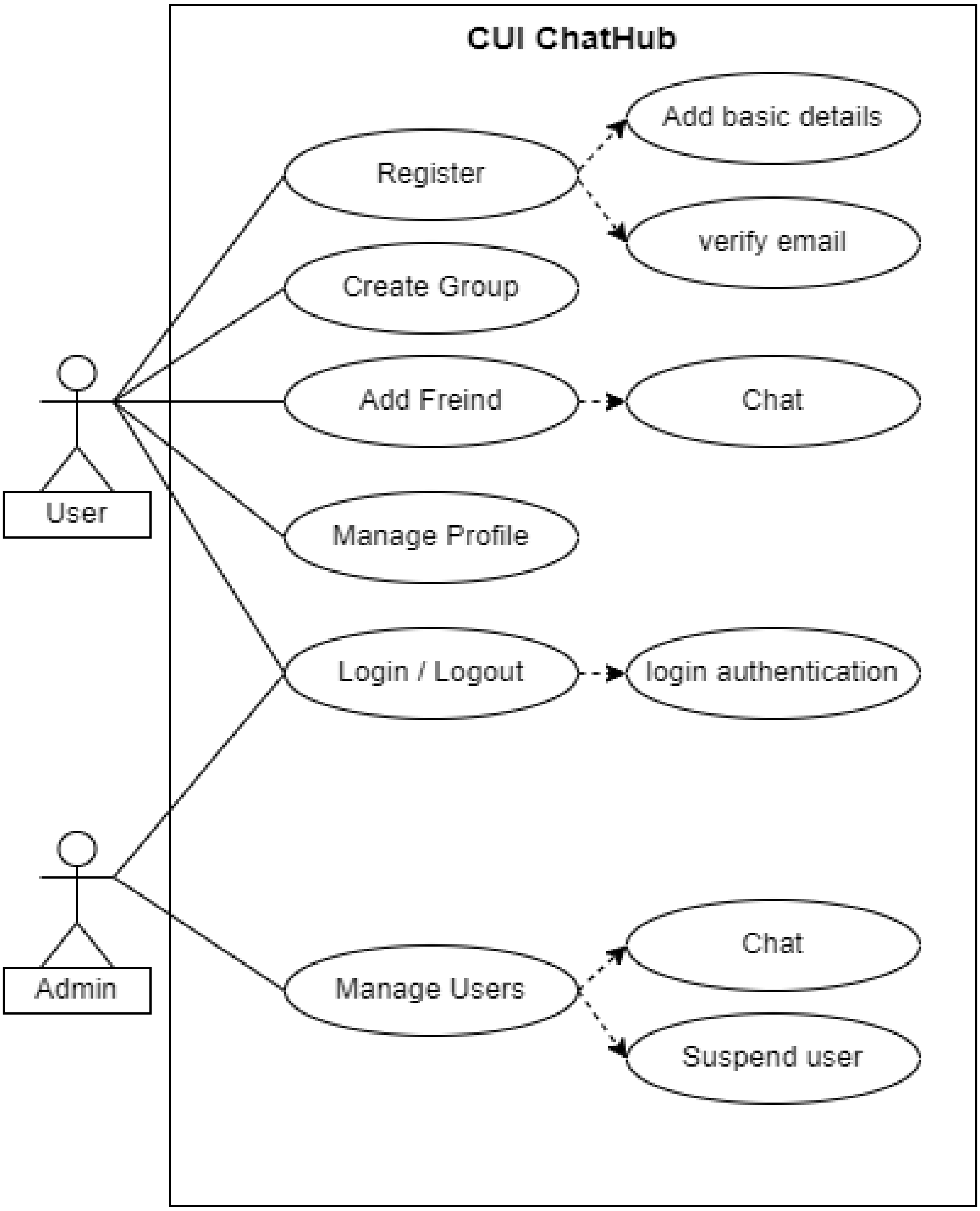

Figure 3.1 CUI Chathub Use Case

3.2 Use Case Descriptions

The table below indicate comprehensive use case descriptions

Table 3.1 Use case description for login/logout

Use Case Id	UC-01
Use Case Name	Login/ Logout
Actors	Primary Actor: User, Admin
Description	The login and logout are blueprint consists of two use cases. The first use case, models the login and the logout procedures; that is, it models how a user is to register as well as deregister as a current user.
Trigger	User and admin want to access the system.
Preconditions	User must be registered in the CUI Chathub.
Post Condition	User can access the system.

Table 3.2 Use case description for user registration

Use Case Id	UC-02
Use Case Name	Register
Actors	Primary Actor: User
Description	Any user willing to use CUI Chathub must first register himself.
Trigger	User indicate he want to register in the app
Preconditions	User must be member of the university and should have valid email address.
Post Condition	User can register himself.

Table 3.3 Use case description for group creation

Use Case Id	UC-03
Use Case Name	Create Group
Actors	Primary Actor: User
Description	While using the CUI Chathub, user of this application can have the privilege to create Group for different purposes and add other users in the group.

Trigger	The user indicates that he wants to create group
Preconditions	User must register in the CUI Chathub.
Post Condition	User can create group.

Table 3.4 Use case description for add friend

Use Case Id	UC-04
Use Case Name	Add Friend
Actors	Primary Actor: User
Description	The user using CUI Chathub can add other users as their friend.
Trigger	User indicate he want to add a friend.
Preconditions	User must have registered himself.
Post Condition	User can add a friend.

Table 3.5 Use case description for chat

Use Case Id	UC-05
Use Case Name	Chat User
Actors	Primary Actor: User
Description	Once user has added another user as his friend, he can start chat with him.
Trigger	User indicate he want to chat with friend.
Preconditions	To chat with the other user, he must have to be in your friend list
Post Condition	User can chat with friend.

Table 3.6 Use case description for profile management

Use Case Id	UC-07
Use Case Name	Manage Profile
Actors	Primary Actor: User
Description	User of the application can Manage his profile. There are certain things which can be

Trigger	User indicate he want manage his profile.
Preconditions	User must have registered himself
Post Condition	User can manage his profile.

Table 3.7 Use case description for user management

Use Case Id	UC-11
Use Case Name	Manage Users
Actors	Primary Actor: Admin
Description	Admin can manage all the users registered within the application.
Trigger	Admin indicate he want to manage the users.
Preconditions	Admin must have linked with the database.
Post Condition	Admin can manage the users.

Table 3.8 Use case description for deleting user

Use Case Id	UC-13
Use Case Name	Delete user
Actors	Primary Actor: User
Description	Admin can suspend any of the user.
Trigger	Admin indicate he want to suspend the user.
Preconditions	Admin must have linked with the database.
Post Condition	Admin can delete the user.

3.3 Functional Requirements

Functional requirements are product features or functions those developers must implement to enable users to accomplish their tasks. Therefore, it is important to make them clear for both the development team and the stakeholders.

This system will have the following modules:

3.3.1 User Registration

User must be able to register for the application through a valid university email address. The user's email address will be the unique identifier of his/her account on CUI Chathub.

3.3.2 Adding friend

A user should see all the users available on the application, so that if user wanted to send friend request to any of them, he can do it.

3.3.3 Send Message

User should be able to send instant message to any of his/her friend

3.3.4 Group Message

User should be able to create groups, and can broadcast messages to these groups.

3.3.5 User Profile Management

Users should be able to edit their profiles.

3.3.6 Group Chat Management

Allow group chat creators to manage group settings, such as adding or removing members.

3.3.7 Emoji Support

Allow users to use a wide range of emoji's in their messages to enhance communication.

3.3.8 Search Friend

Implement a robust search and filtering system for users to easily find friend

3.4 Nonfunctional Requirements

Nonfunctional requirements are those that establish parameters that may be used to evaluatethe performance of a system. Nonfunctional requirements are the limitations to which the system will work.

3.4.1 Performance

The system should be able to handle a large number of concurrent users and should provide fast response times for user actions

3.4.2 Security

The system should have robust security measures in place to protect user credentials and prevent unauthorized access.

3.4.3 Availability

This system must be always accessible, 24 hours a day, seven days a week.

3.4.4 Ease of Use

A basic yet high-quality user interface should be created to make it simple for the users to use the application.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1 System Architecture

The structure of the system explains the working of the system and describes the components of the CUI Chathub, their relationships, and how they interact with each other, with this architecture. [4]

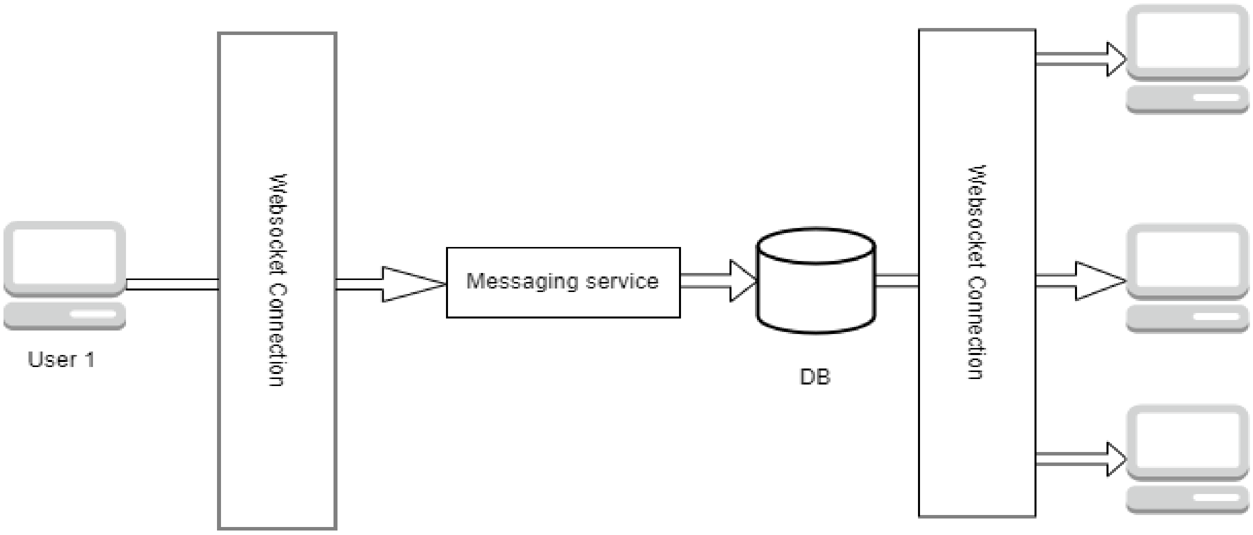

Figure 4.1CUI Chathub chat architecture

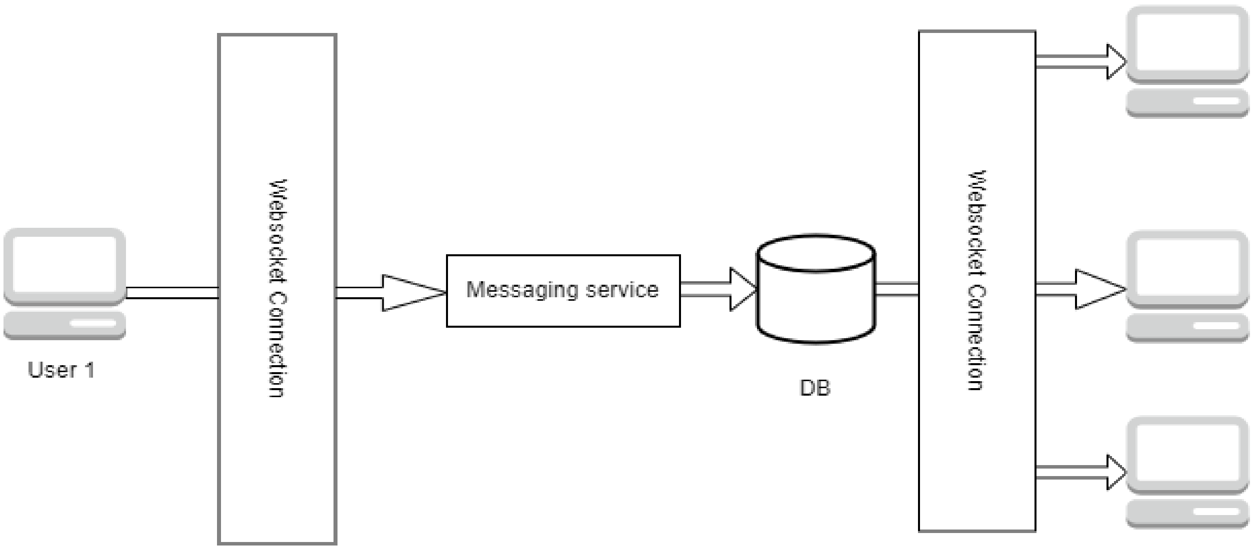

Figure 4.2 CUI Chathub Group Chat Architecture

4.2 Process Flow/Representation

The Process Flow Representation illustrates the step-by-step sequence of actions and interactions between different components in the CUI Chathub. It provides a visual overview of how users interact with the platform to perform various tasks. [5]

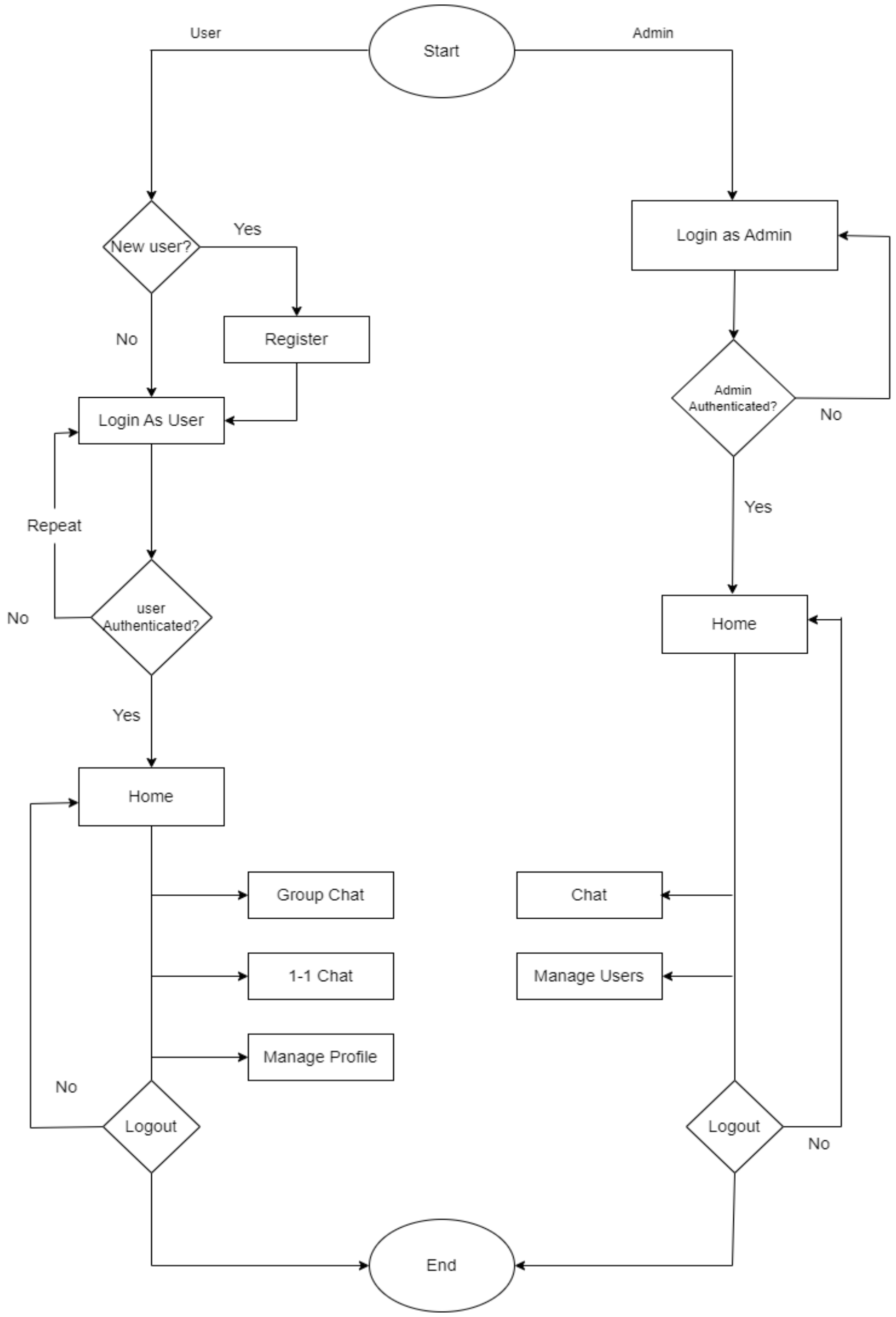

Figure 4.3 CUI Chathub Process Flow

4.3 Sequence Diagram

These figure shows how the system will work through sequence diagrams, task after task sequence wise. It gives a complete description of the working of CUI Chathub. [6]

Phase 1: Authentication

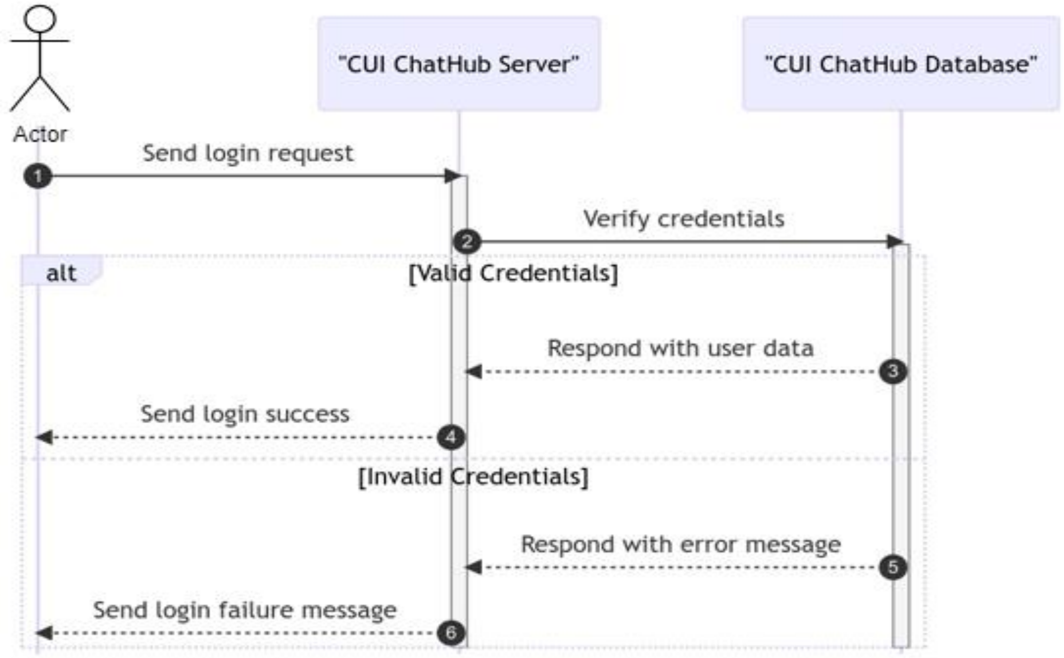

Figure 4.4 CUI Chathub Authentication Sequence Diagram

Phase 2: Chat Sequence

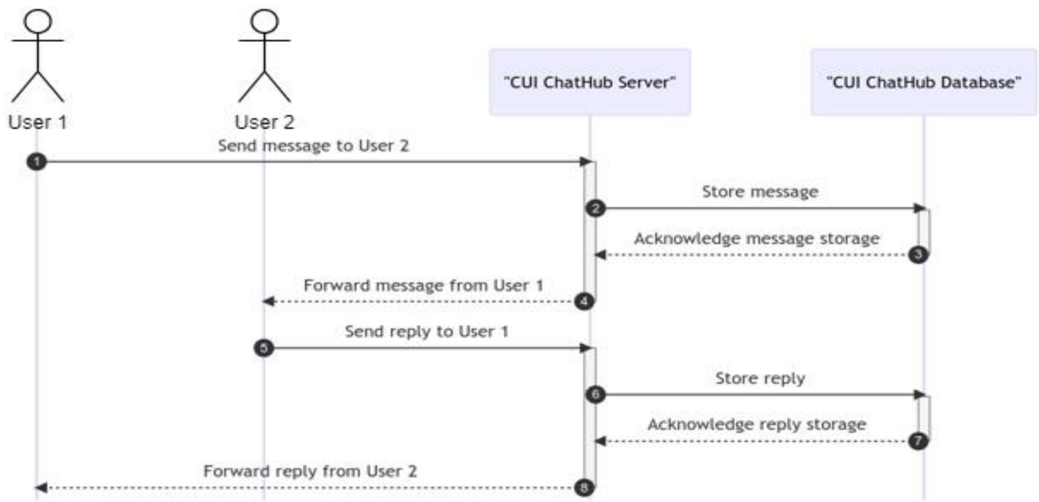

Figure 4.5 CUI Chathub Sequence diagram

4.4 Class Diagram

This class diagram is a graphical representation of the system's classes, their attributes, and the relationships between them. It elaborates structure of the CUI Chathub and shows how the different classes in the system are related to each other.

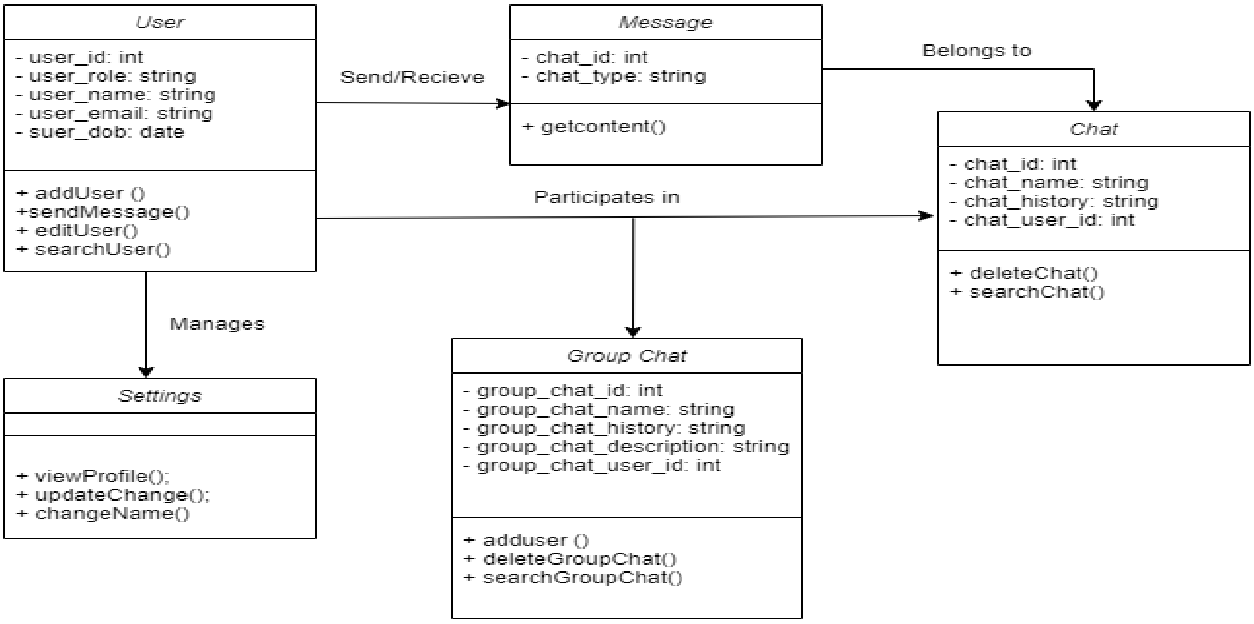

Figure 4.6 CUI Chathub Class Diagram

Chapter 5

Implementation

5 Implementation

In the implementation phase of CUI Chathub, we have devised algorithms to govern the application's core functionalities, fostering seamless interactions among university students, societies, and administrators.

5.1 JWT-Based User Authentication

CUI Chathub uses JWT for secure user authentication, encoding ID and roles. This enhances security and scalability by eliminating continuous session tracking. [7]

5.2 User Registration Algorithm

This algorithm manages the user registration process, enabling new users to join CUI Chathub with ease. It ensures the secure storage of user data and the streamlined approval of memberships by administrators.

5.3 Real Time Chat and Messaging Algorithm

Central to CUI Chathub's functionality, Real-time communication is emphasized through a WebSocket module, enabling instant messaging and dynamic updates for interactive chat sessions. This algorithm governs user-to-user and group messaging features. Users can initiate one-on-one or group chats, fostering real-time communication within the university community. [8]

5.4 Security and Data Protection

Throughout the implementation, importance is given to security. Robust encryption techniques safeguards sensitive information.

5.5 User interface

Below figures represent the user interfaces and screens of CUI qChathub. Figure 5.1shows Log in Screen of System, Figure 5.2 show Sign up screen, Figure 5.3 shows Home Screen of application Figure 5.4 shows Friends and contact list and Figure 5.5 show chat one to one with user screen.

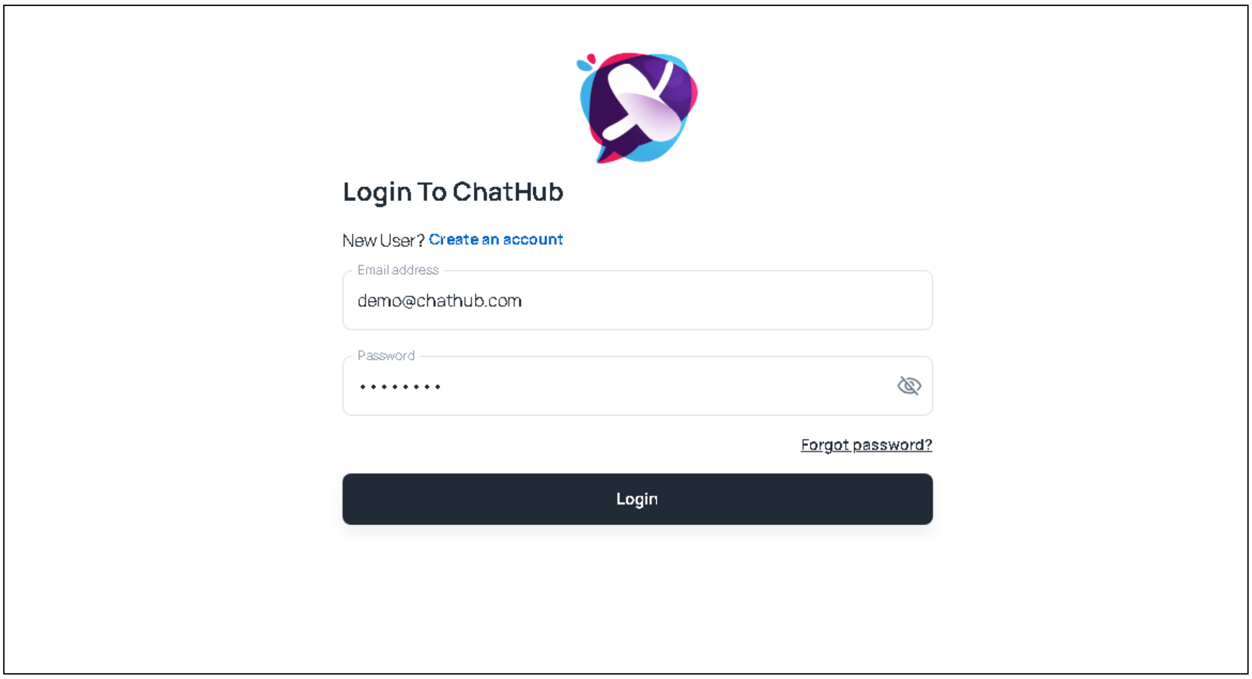

Figure 5.1 CUI Chathub Log in Screen

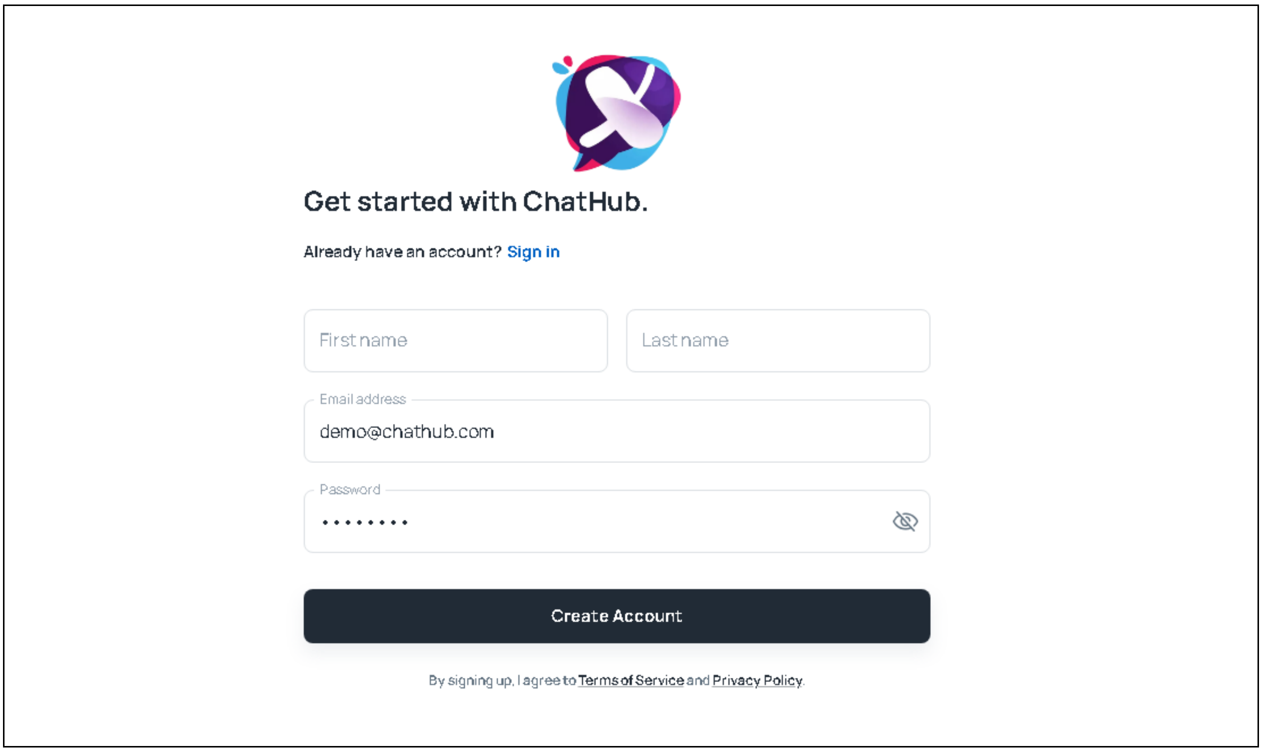

Figure 5.2 CUI Chathub Sign Up Screen

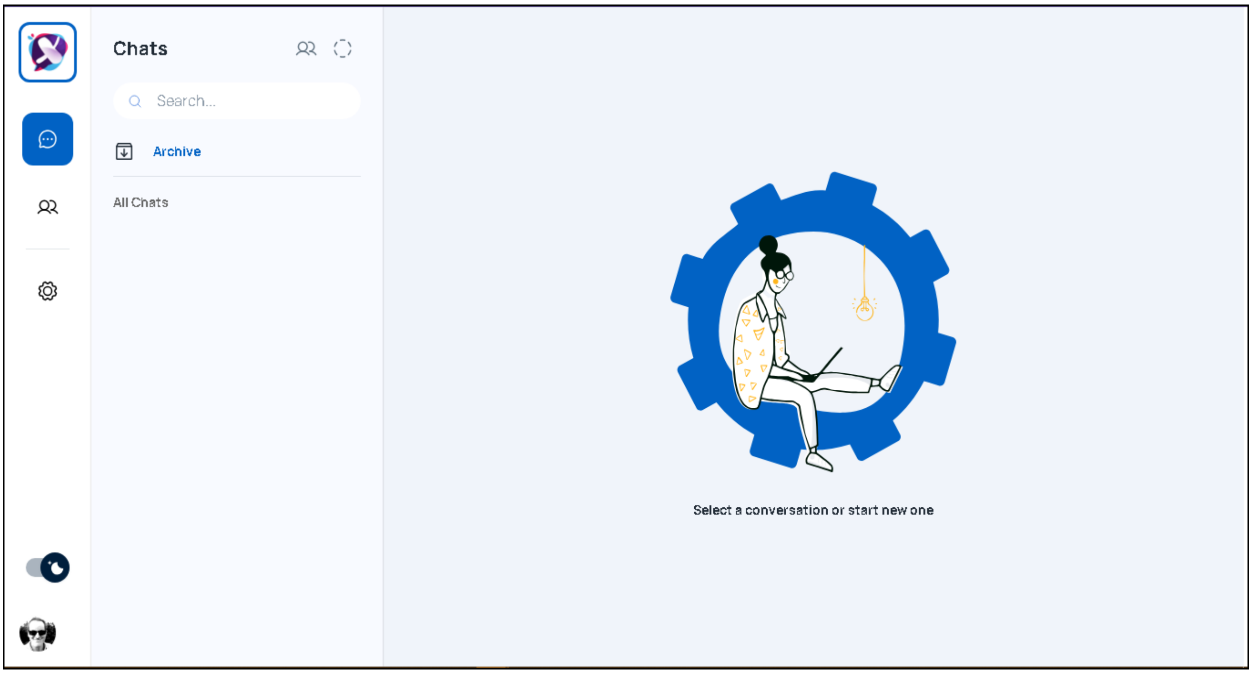

Figure 5.3 CUI Chathub Home Screen

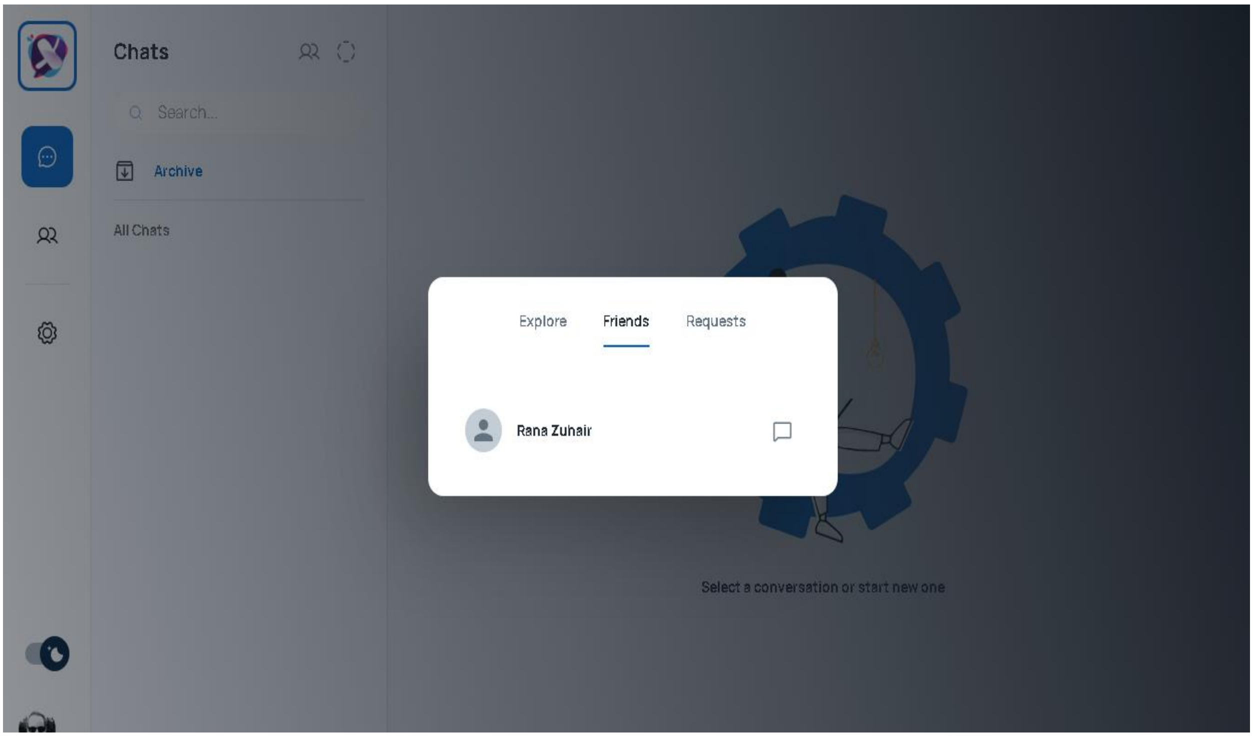

Figure 5.4 CUI Chathub Friends Screen

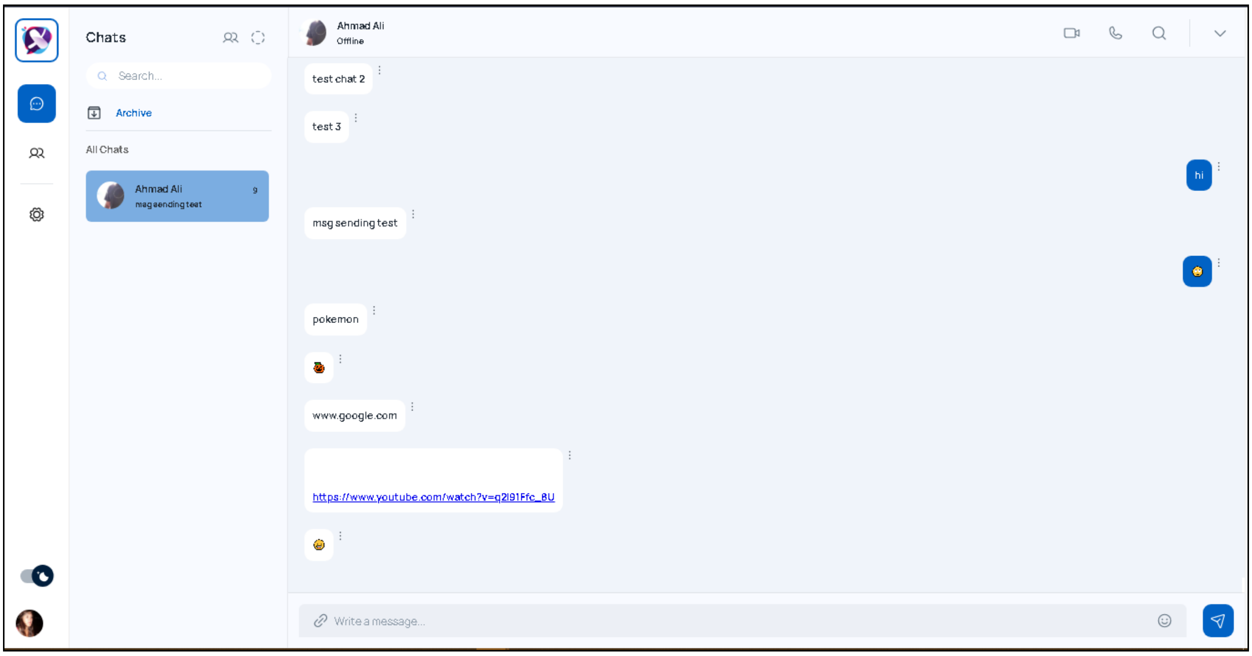

Figure 5.5 CUI Chathub Chatting Screen

Chapter 6

Testing and Evaluation

6 Testing

6.1 Manual Testing

During the manual testing phase, we rigorously assessed "CUI Chathub" for various functionalities. Any issues identified during testing were promptly addressed to guarantee a seamless user experience within the application.

Table 6.1 Manual Testing

Test Cases	Expected Result	Actual Result	Pass/Fail
App Launch	Successful	Successful	Pass
User Registration	Successful	Successful	Pass
User Login	Successful	Successful	Pass
Home Page Display	Accurate	Accurate	Pass
Search User	Accurate	Accurate	Pass
Add User	Accurate	Accurate	Pass
Send Message	Accurate	Accurate	Pass
Create Group	Successful	Successful	Pass
Manage Profile	Successful	Successful	Pass
App Performance and Response	Smooth	Smooth	Pass
App Navigation	Easy	Easy	Pass
App Usability and User Friendliness	High	High	Pass
Logout	Successful	Successful	Pass

6.2 Unit Testing

Unit tests is conducted to assess the individual components of the CUI Chathub.

Table 6.2 Unit Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	User Registration	Valid user information	User account is created successfully	Pass
UT-002	User Authentication	Valid user credentials	Successful authentication and user login	Pass
UT-003	Blank Password Field	Leave the password field blank during registration	System displays an error message	Pass

UT-004	Chat Message Sending	Sending a message	Message is sent	Pass
UT-005	Group Creation	Creating a new group	Group is created, and the creator is a member	Pass
UT-006	Group Joining	Joining an existing group	User becomes a member of the group	Pass

6.3 Functional Testing

Functional testing verifies End to end functionalities of a CUI Chathub against its functional requirements and specifications, to ensure that the software behaves as expected.

Table 6.3 Functional Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	User Registration	Registering with valid university email address	User accounts can be created with valid email addresses	Pass
Ft-002	Searching user	Searching user in the search field	If user exists, used should be displayed	Pass
FT-003	Real-Time Chat	Sending and receiving messages in real-time	Messages are sent and received successfully	Pass
FT-004	Group Chat Creation	Creating group chats	Group chats can be created and messages sent to groups	Pass
FT-005	User Profile Management	Editing user profiles	User profiles can be edited and updated	Pass
FT-006	Password Encryption	Password encryption	Passwords are securely encrypted a	Pass

6.4 Integration Testing

Integration testing is a software testing technique that focuses on evaluating the interactions between different software modules or components of CUI Chathub

Table 6.4 Integration testing

Test Case ID	**Test Case Name**	**Attributes & Values Tested**	**Expected Result**	**Actual Result**
IT-001	Test and configure Websocket	Webscoket realtime conenction	Connection Established	Pass
IT-002	Test and configure. the database.	Databse realtime connection	Connection established	Pass

Chapter 7

Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The CUI Chathub project represents a significant step forward in enhancing communication and collaboration within the university community. The project's development and implementation have addressed several challenges that were prevalent in the existing system, providing a specialized, user-centric, and adaptable platform for students, faculty, and university organizations.

The system's user-friendly interface and real-time chat functionality have fostered a more engaged and connected community, allowing students to easily form groups, join discussions, and share information. The platform has also empowered university societies and clubs to efficiently share announcements, events, and updates with their members, leading to improved participation and awareness.

As we conclude this project, it is clear that CUI Chathub has met its objectives in providing a dedicated and agile communication tool that aligns with the unique requirements of a university setting. The system's flexibility, adaptability, and focus on user experience have significantly enhanced the overall university experience for its users.

7.2 Future Work

While the initial development of CUI Chathub has been successful, there is always room for improvement and expansion. Some areas for future work and enhancement include:

- **Call and Media Sharing:**
  Continuously adding new features and capabilities to meet evolving user needs, such as voice call and media sharing,
- **Enhanced Security:**
  Strengthening the security features to ensure the protection of user data and privacy
- **Scalability:**
  Optimizing the system's architecture to accommodate a growing user base and increased data volume without compromising performance.
- **User Feedback and Iteration:**
  Regularly gathering user feedback and using it to make iterative improvements to the platform.
- **Mobile Application Development:**
  Ensuring the mobile applications (iOS and Android) are available for the users to use.

- **User Training and Support:**

  Expanding user training materials and support channels to help users make the most of the platform. Incorporating these future work elements will help CUI Chathub stay current, relevant, and effective in catering to the dynamic needs of the university community while maintaining a high standard of security and usability.

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